



Welcome to the ERP

Project Market!

NP: Hai Wang h.wang5@uu.nl;

SCMB: Marijn van Huis m.a.vanhuis@uu.nl

IMAU: Alexis Gilbert a.r.gilbert@uu.nl

Jl: Peter van Capel p.j.s.vancapel@uu.nl

GRASP: Marta Verweij m.verweij@uu.nl

Targets

Week 3-4

Question



Week 4-8

Implementation:
setting up,
measurments...



Week 7-9

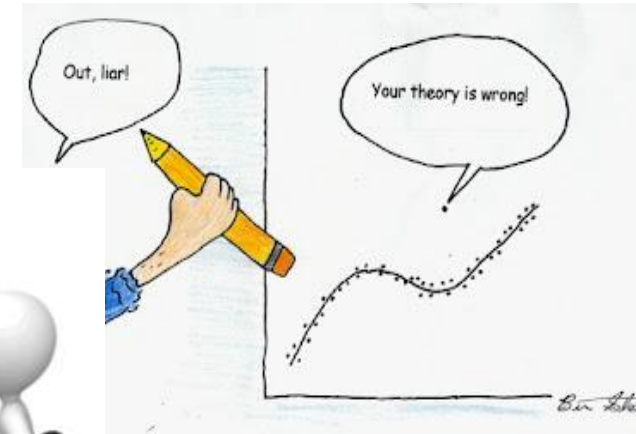
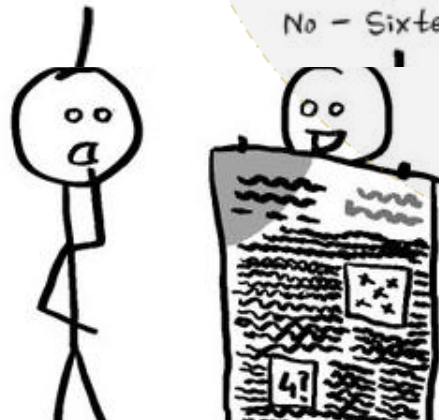
Interpretation:
analysis &
modelling



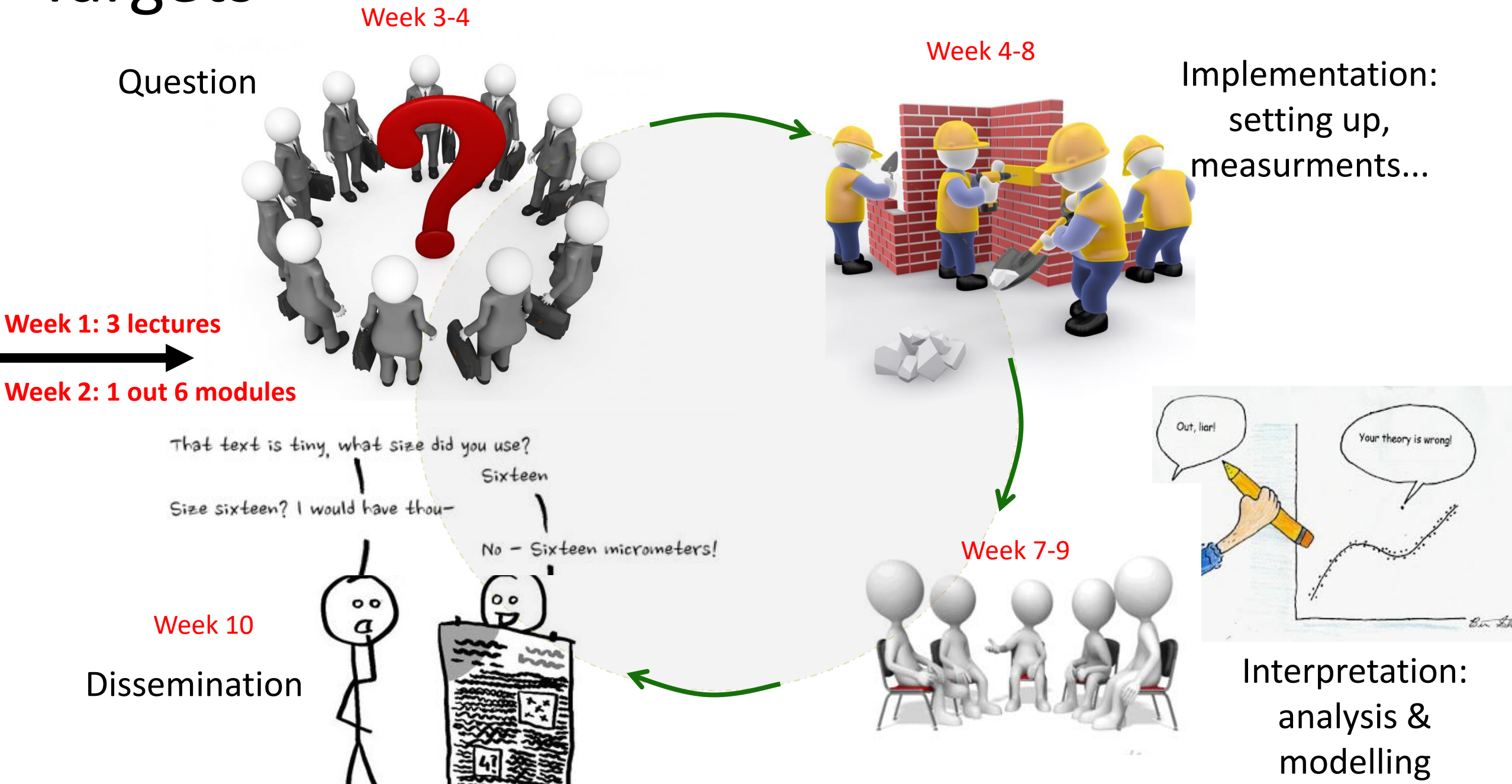
Week 10

Dissemination

That text is tiny, what size did you use?
Sixteen
Size sixteen? I would have thou-
No - Sixteen micrometers!



Targets





ERP All-in-One Document

Welcome.

This all-in-one document includes all information for teachers, students, and teaching assistants (TAs) for experimental research project, 2026. Please send an email to Hai Wang (h.wang5@uu.nl) and cc Marijn van Huis (M.A.vanHuis@uu.nl) if something is unclear or missing. We will keep the documents updated while do our best to keep the structure logical and clear.



1. Introduction to the course and schedule

2. Learning goals and objects

3. Implementation: When/where who to do what -guides for students/teacher/TAs

3.1 Guides to students

3.2 Guides to TAs

3.3 Guides to teachers:

4. Information for general lectures and Modules in week 1-2

5. Project description (for week 3-9)

5.1 A quick overview on the teacher and project (to be updated for 2026)

5.2 Project description

6. Group Project Proposal Handout

7. Rubric for evaluation



When	Where	What/Who
Week 0	Online; A link will be provided	Assignment for students: Please form a group of 3 students yourself latest at 13:00, on 21/04. We will help you form a group whenever necessary.
Week 1: 22.04; 09:00-12:00	Minnaert. 001, 002, 003, 004	Lecture 1: project information and project matching Coordinators will provide an overview of the projects that students will work on; then students will break out in 3 lecture
		room, and have the chance to talk to teacher representatives (at least one per institute) who can tell students more details on the project
Week 1: 22-26/04	Online	Assignment for students: Submitting your 5 wishes for the projects that your group would like to work on; a link will be provided on 22/04 after lecture 1. You will be randomly assigned to a project if you fail to submit your choices in time
Week 1: 22.04; 13:15-15:00	KBG Atlas	Lecture 2: Introduction to the course (13:15-15:00; 22.04) Coordinators will present the course structure, schedule, expectation, and implementation to students; Students will get the chance to ask questions to the course coordinators.
Week 1: 24.04; 09-10:45	KBG Atlas	Lecture 3 : Collaboration and team work (09:00-10:45; 24.04) Dr. Ria Dolfine will discuss and reflect together how to work together as a group. The assignment with the deadline on 15 th of May is related to this lecture -so it is important that you and your teammates can join it together.



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**5. Project description (for week 3-9; 47 projects in 2026 in total)****5.1 A quick overview on the teacher and project (to be updated for 2026)**

22-4-2026

Note that: project 19 and 25 are open projects. If you would like to develop a new topic of yourselves, that is possible and please talk to Hai Wang/Marijn van Huis. Please put these two projects as your top 2 wishes if you would like to do an open project.

#	Teachers	Project name
1	Marc Schneider, Erik van Seville	<u>Measuring water surface movements with a self-built floating GNSS tracking system</u>
2	Marc Schneider, Erik van Seville	<u>Data collection and analysis for an outdoor GNSS tracking system</u>
3	Carleen Tijm-Reijmer	Meteorological measurements: Temperature, humidity and wind
4	Carleen Tijm-Reijmer	Meteorological measurements: Solar UV/VIS/IR radiation and wind measurements
5	Carleen Tijm-Reijmer	Meteorological measurements: Solar UV/VIS/IR radiation and solar energy production
6	Alexis Gilbert, Thomas Röckmann	Isotopic composition of methane (CH ₄) emitted from different sources
7	Alexis Gilbert, Thomas Röckmann	Natural isotopic composition of carbon dioxide (CO ₂) emitted from different sources
8	Alexis Gilbert, Thomas Röckmann	Isotopic fractionation of carbon monoxide (CO) in photolytic simulation experiments
9	Rupert Holzinger & Nemat Omidikia	Environmental Research Using a Very Sensitive and Specific Mass Spectrometer I
10	Rupert Holzinger & Nemat Omidikia	Environmental Research Using a Very Sensitive and Specific Mass Spectrometer II
11	Allard Mosk	<u>The electro-optics cookbook</u>
12	Allard Mosk	<u>I spy with my little lensless camera</u>



Nanophotonics research projects

Contact: **Hai Wang, h.wang5@uu.nl**



Utrecht University

DEBYE INSTITUTE IN PHYSICS

Nanophotonics: who we are

Study optics and ultrafast dynamics, including topics like light-matter interactions, imaging and dynamics in quantum materials



Light matters!

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a.p.mosk@uu.nl



Sanli Faez,
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Dries van Oosten
d.vanoosten@uu.nl



Peter van der Straten
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Hai Wang
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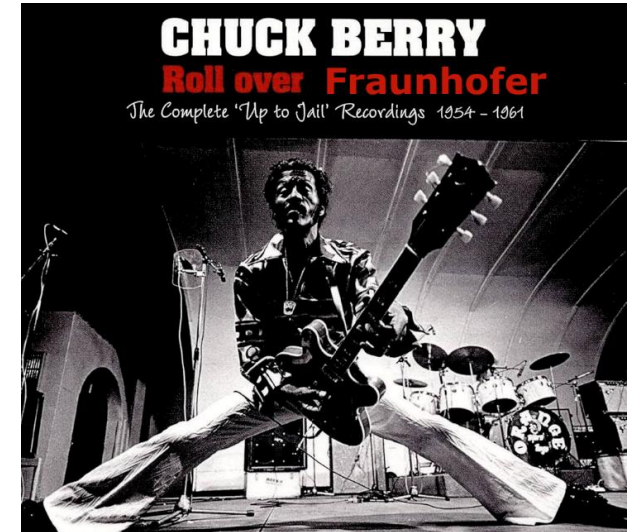




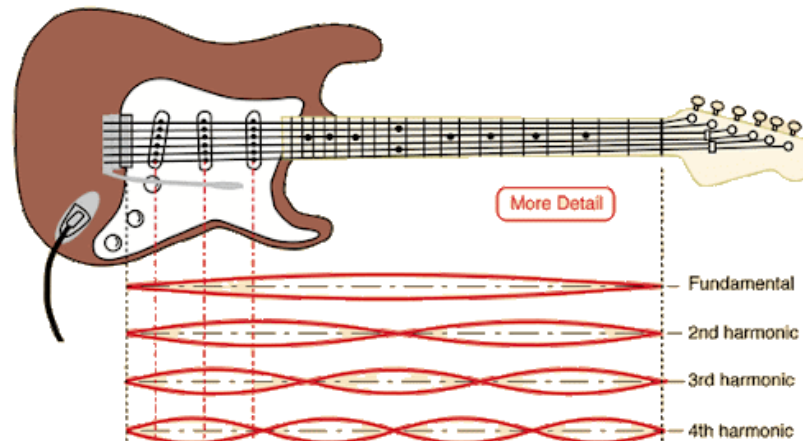
Project 12-16 (optical/sound) waves

The Physics of Music

(1) *How does a MP3 work? (13)*



(2) The physics of musical instruments (16)

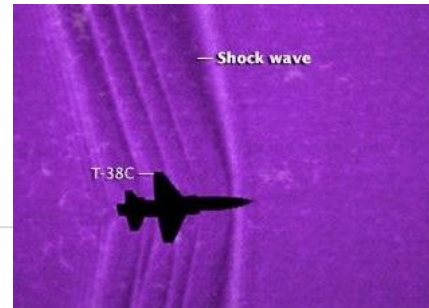




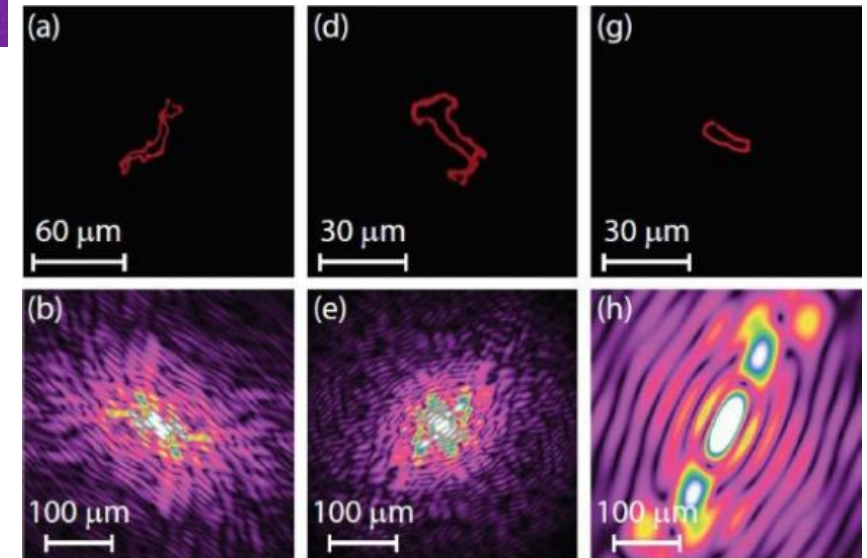
Project 12-16 (optical/sound) waves

(1) Acoustic experiments: ranging and detection of objects (sonar) (14)

(2) Seeing shock wave (15)



(3) I spy without a camera (12)

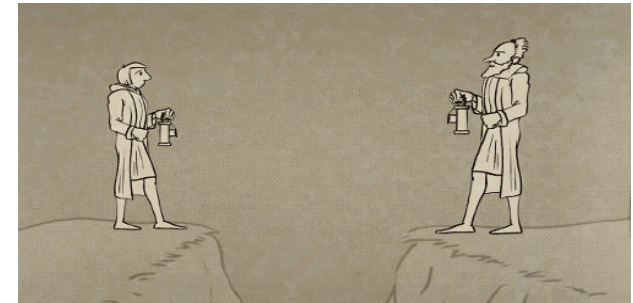




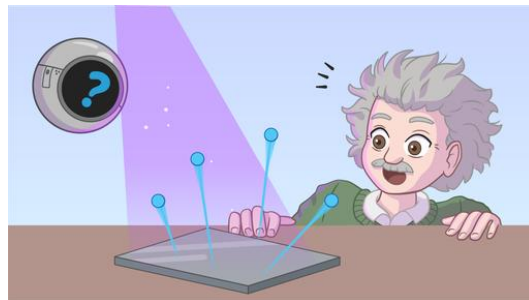
Project 11, 17–20: optical spectroscopy

Nature constant measurements:

(1) *Speed of light (16)*



(2) *Planck constant by photoelectrical effects (19) or cooking electronics (11)*

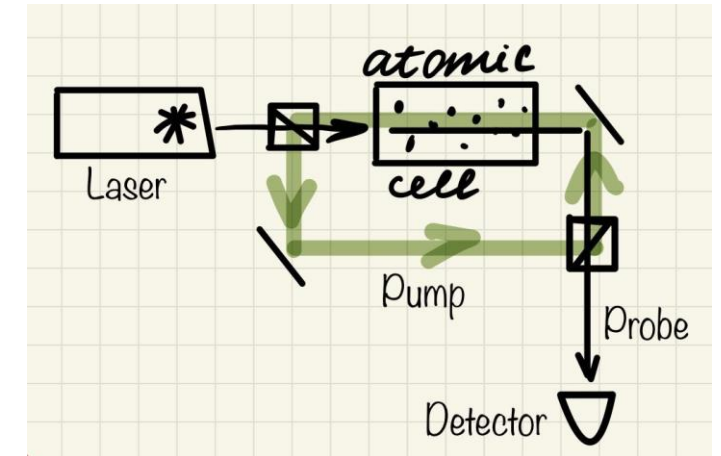




Project 17-20: optical spectroscopy

Laser spectroscopies

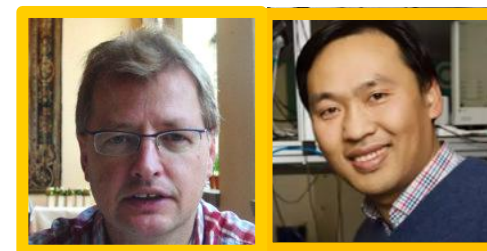
(1) Saturation spectroscopy



(2)



Optical identification of a “grey” film by Raman spectroscopy





Soft Condensed Matter and Biophysics research projects

Contact: **Marijn van Huis,**
m.a.vanhuis@uu.nl



Soft Condensed Matter and Biophysics

We study the **physics of soft matter and biological model systems** and, using these insights, **fabricate new mesostructured materials** from them.

Our research tools consist of:

- confocal microscopy
- electron microscopy
- scattering methods
- colloid synthesis
- computer simulations and theory





Projects 35–39: ordering of soft matter

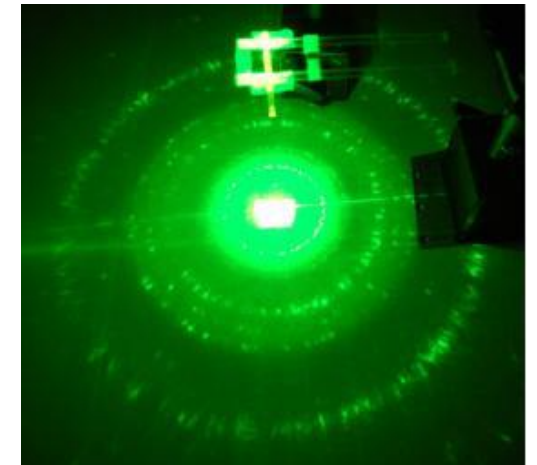
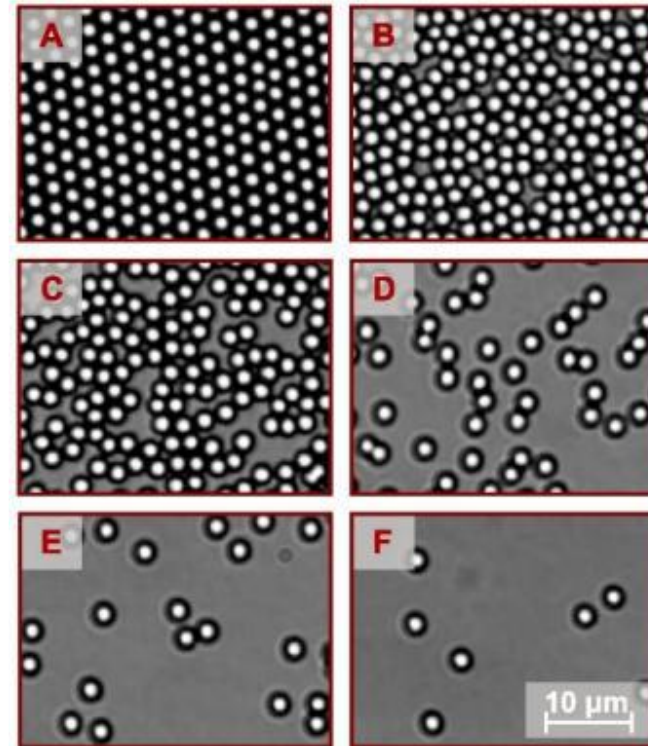
*Crystallography of light
(Self)organisation of particles*



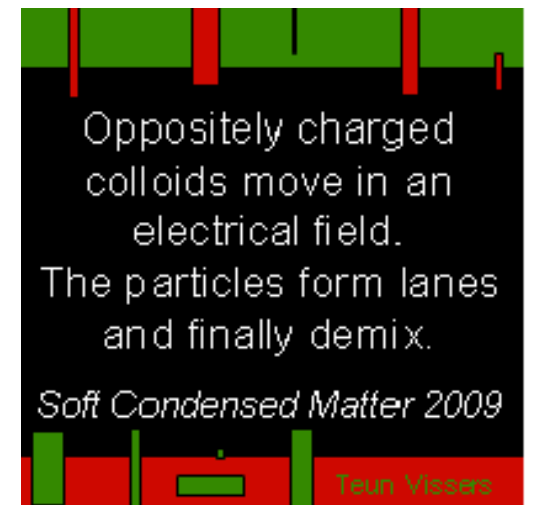
Dr. Arnout Imhof



Prof. Alfons van Blaaderen

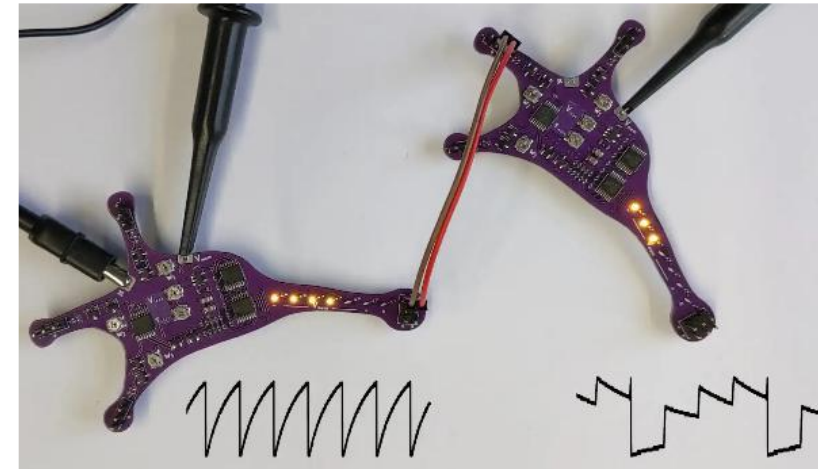
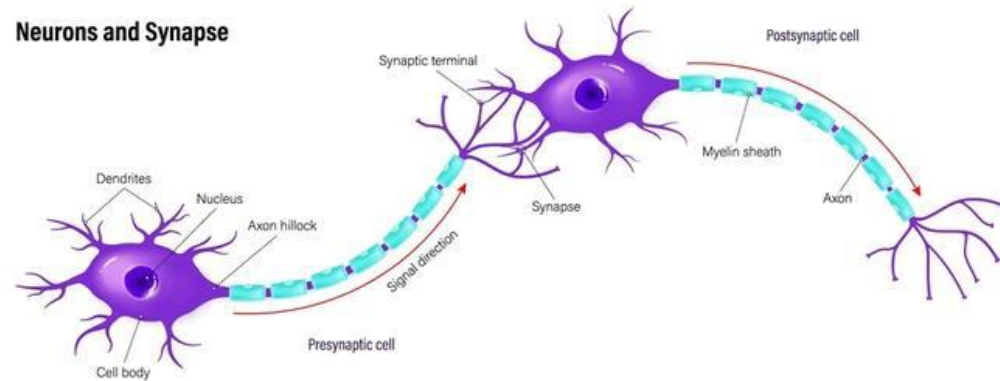


*Measuring pair potentials between particles
Phase behaviour of 2D colloidal systems*





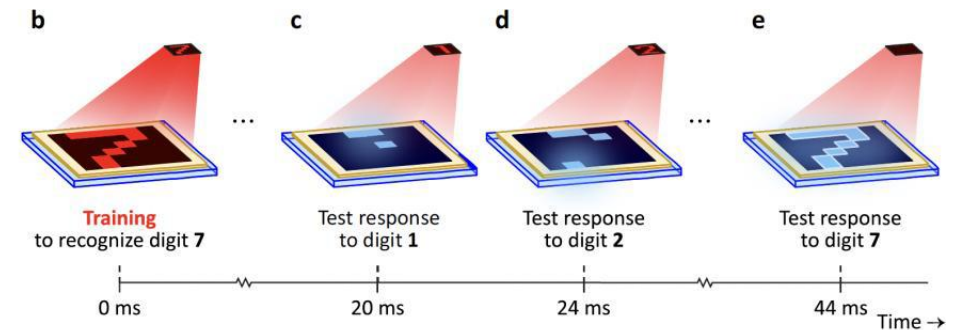
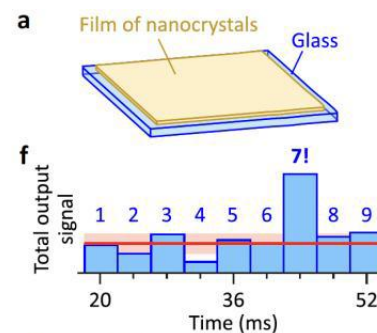
Projects 33, 42, 43: Neuromorphic computing



Prof. Marjolein Dijkstra



Dr. Freddy Rabouw



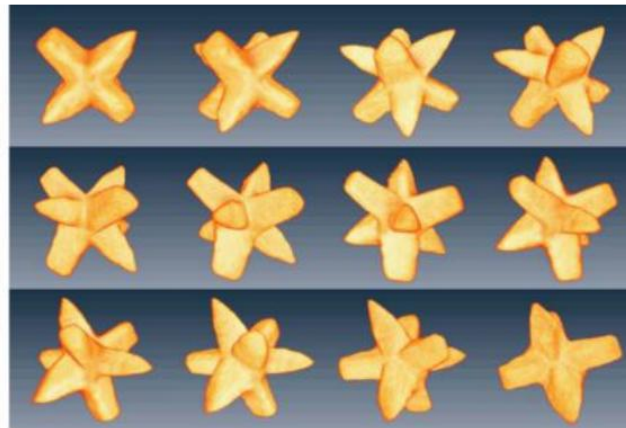
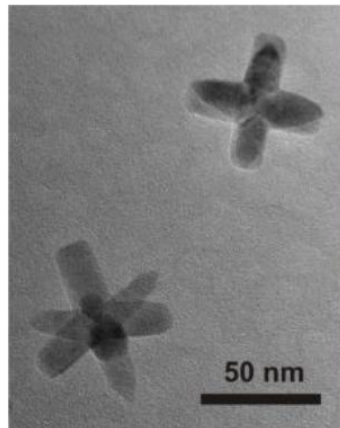


Projects 34, 40, 41: *Tomographic reconstruction methods*



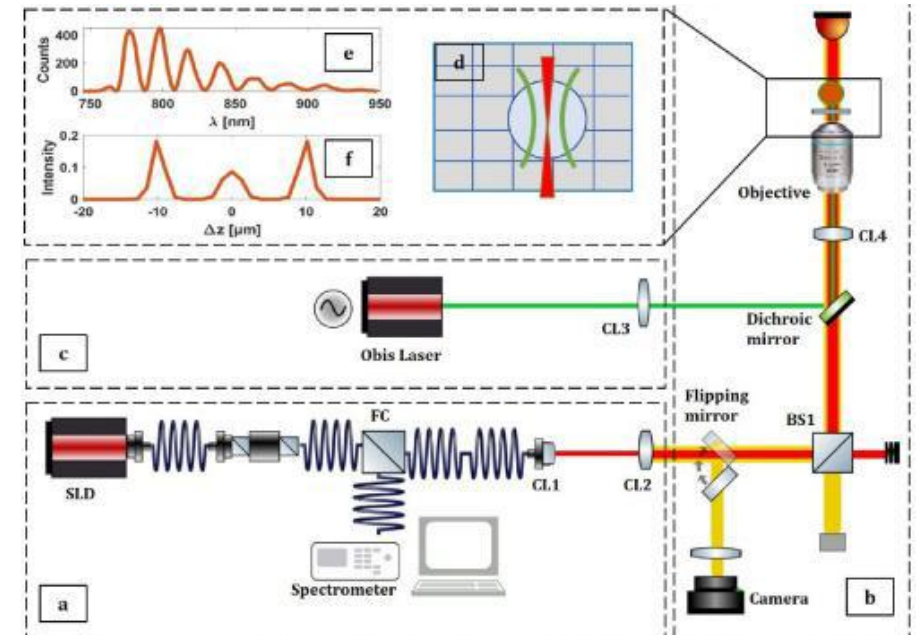
Prof. Marijn van Huis

Electron tomography
of nanoparticles



Dr. Gerhard Blab

Optical
coherence
tomography





IMAU research projects

Contact: **Alexis Gilbert, a.r.gilbert@uu.nl**

IMAU: who we are

The primary mission of the **Institute for Marine and Atmospheric research Utrecht (IMAU)** is to perform high quality fundamental climate research and teaching so as to contribute to solving grand societal challenges related to the global climate crisis

10 projects divided in 4 types



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Carleen Tijm-Reijmer
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Rupert Hotzinger
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Alexis Gilbert
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Projects 1 & 2: Build your own GNSS tracking system!

1) Measuring water surface movements with a self-built floating GNSS tracking system

2) Data collection and analysis for an outdoor GNSS tracking system



Erik van Sebille
e.vansebille@uu.nl

Conception of GNSS tracking system + outdoor testing!



Projects 3 - 5: Build your own weather station!

3) Meteorological measurements: Temperature, humidity and wind

4) Meteorological measurements: Solar UV/VIS/IR radiation and wind measurements

5) Meteorological measurements: Solar UV/VIS/IR radiation and solar energy production

= develop a simple weather station



Carleen Tijm-Reijmer
c.h.tijm-reijmer@uu.nl

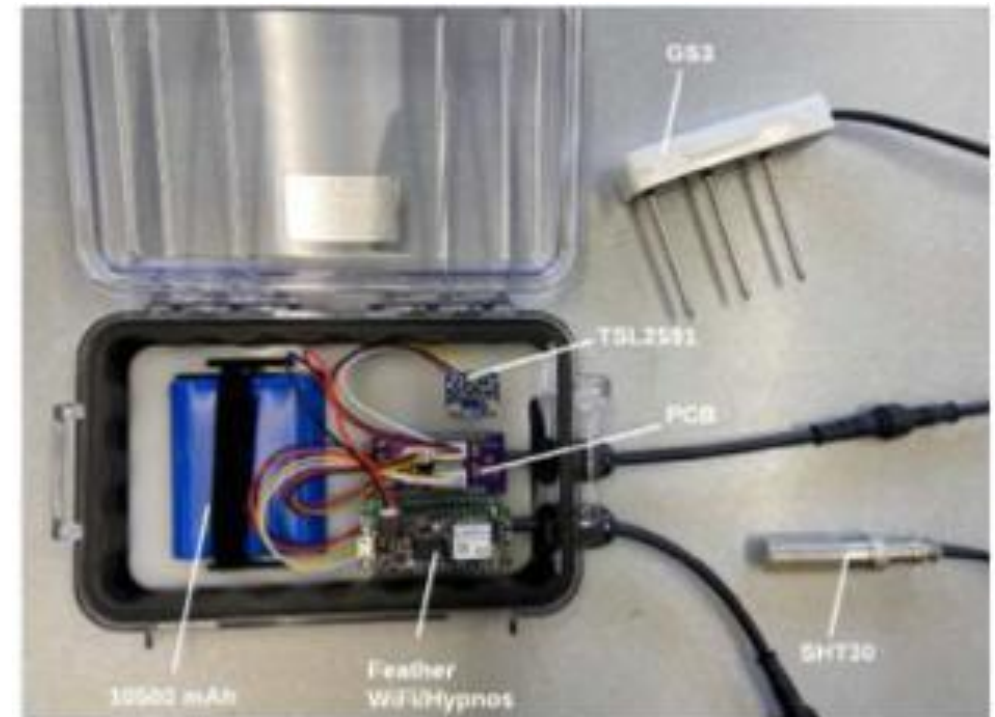


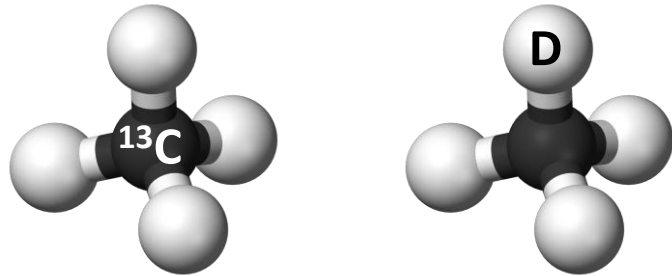
Fig. 1. Completed WeatherChimes device.

Projects 6 - 8: Trace greenhouse gases

- 6) Isotopic composition of methane (CH_4) emitted from different sources
- 7) Natural isotopic composition of carbon dioxide (CO_2) emitted from different sources
- 8) Isotopic fractionation of carbon monoxide (CO) in photolytic simulation experiments



Alexis Gilbert
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Go out and take samples!
Simulate processes in the lab!



Projects 9 & 10: Trace nanoplastics in the environment!

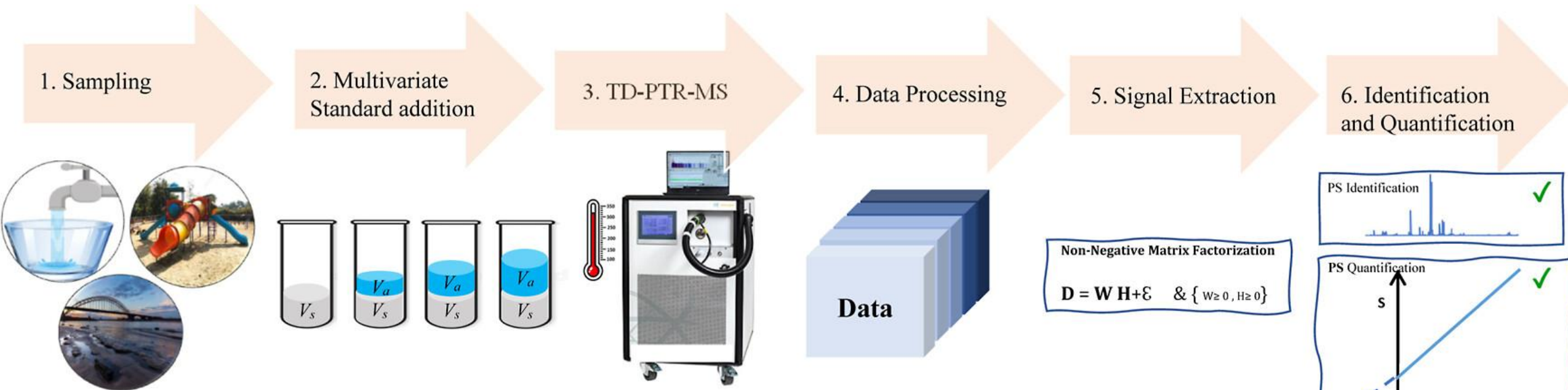
9) Environmental Research Using a Very Sensitive and Specific Mass Spectrometer I

10) Environmental Research Using a Very Sensitive and Specific Mass Spectrometer II



Rupert Hotlzinger
r.holzinger@uu.nl

Use mass spectrometry to measure nanoplastics





GRASP research projects

Contact: Marta Verweij, m.Verweij@uu.nl



GRASP: who we are

Study gravitational waves and particle physics

Chris van den Broeck,
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Thomas Peitzmann
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Marta Verweij
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Anuradha Samajdar
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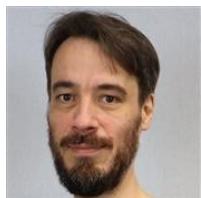
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Marc van der Sluijs
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Alessandro Grelli
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Engineers



Rene Bartel

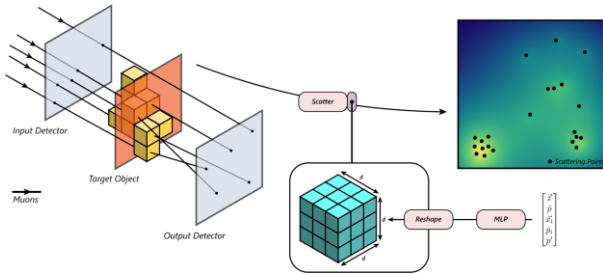


Marcel Rossewicz



Project 21, 22, 23, 26: particle physics

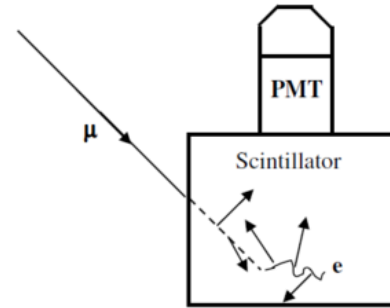
Project 21. Muon Tomography



Can you measure the density of the object in between the scintillators?
Technology: scintillators + robotic scanner

Contact: Marta Verweij, m.verweij@uu.nl

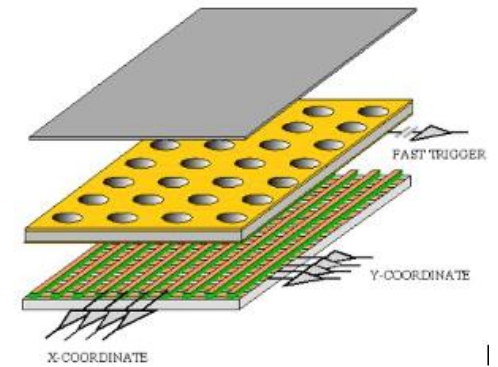
Project 22. Muon Lifetime



Determine the lifetime of muons
by measuring the decay chain of the muons
Technology: Scintillator + photomultiplier

Contact: Thomas Peitzmann, t.peitzmann@uu.nl

Project 23. Muon Tracking with GEMs



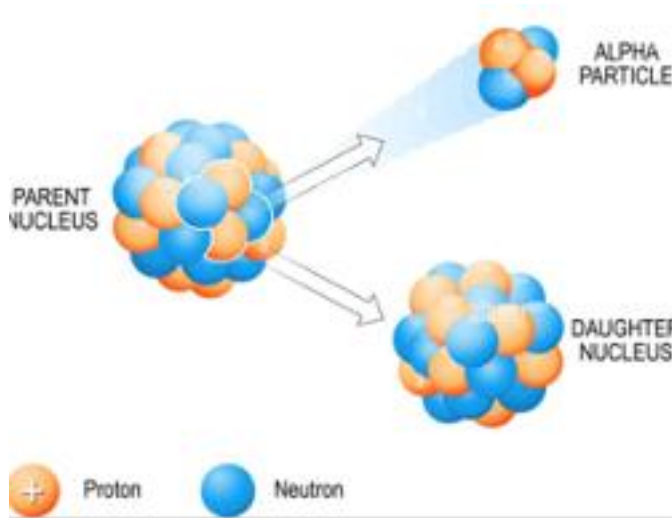
Track the path muons travel
Technology: gas detector

Contact: Alessandro Grelli, a.grelli@uu.nl



Project 21, 22, 23, 26: particle physics

Project 26. Build your own particle detector



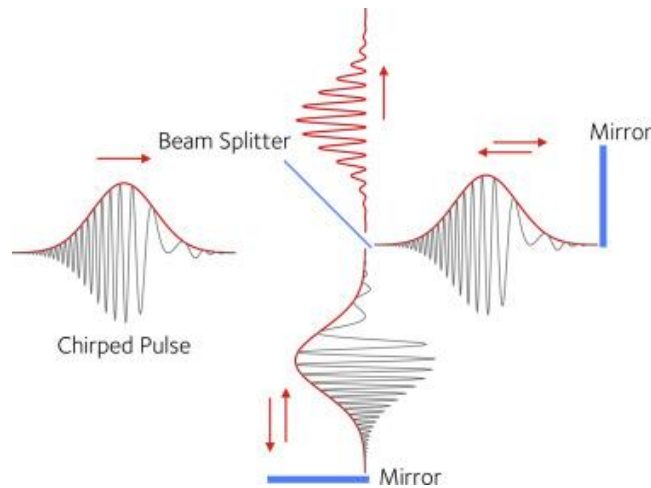
Can you identify different types of radioactive sources using your pocket-size particle detector?

Technology: DIY spectrometer

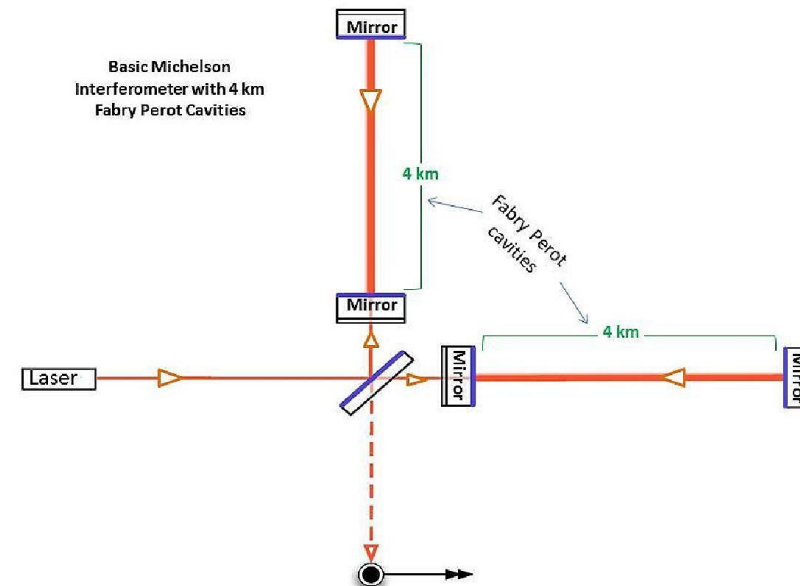


Project 25: Gravitational Waves

Project 26. Thermal Expansion with an Interferometer



Build your own interferometer to detect minute length changes by observing fringe shifts with a photodiode
Technology: Interferometer + laser





Jl / ITP research projects

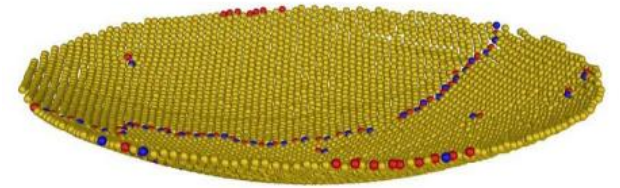
Contact: **Peter van Capel,**
p.j.s.vancapel@uu.nl



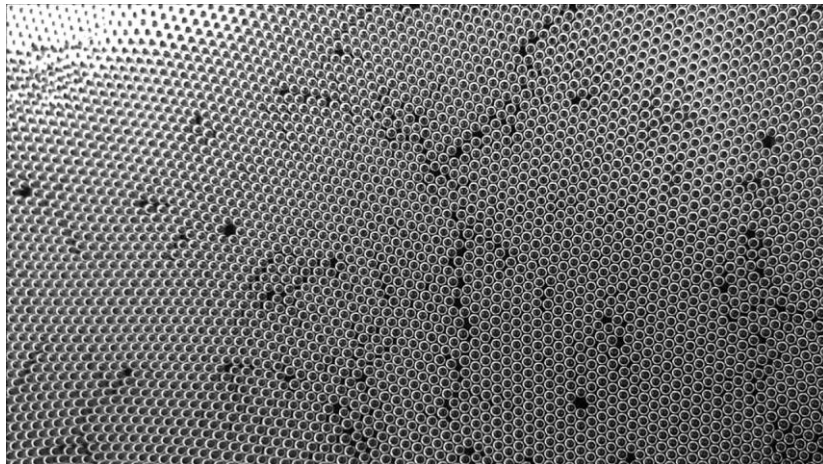
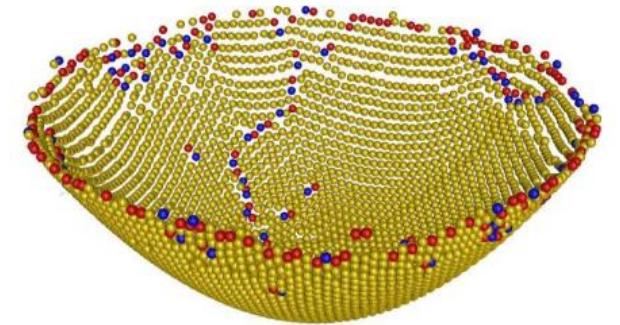
Project 27: Defects in Dense Suspension

Bragg's Bubble Rafts

(1) Shear or deformation

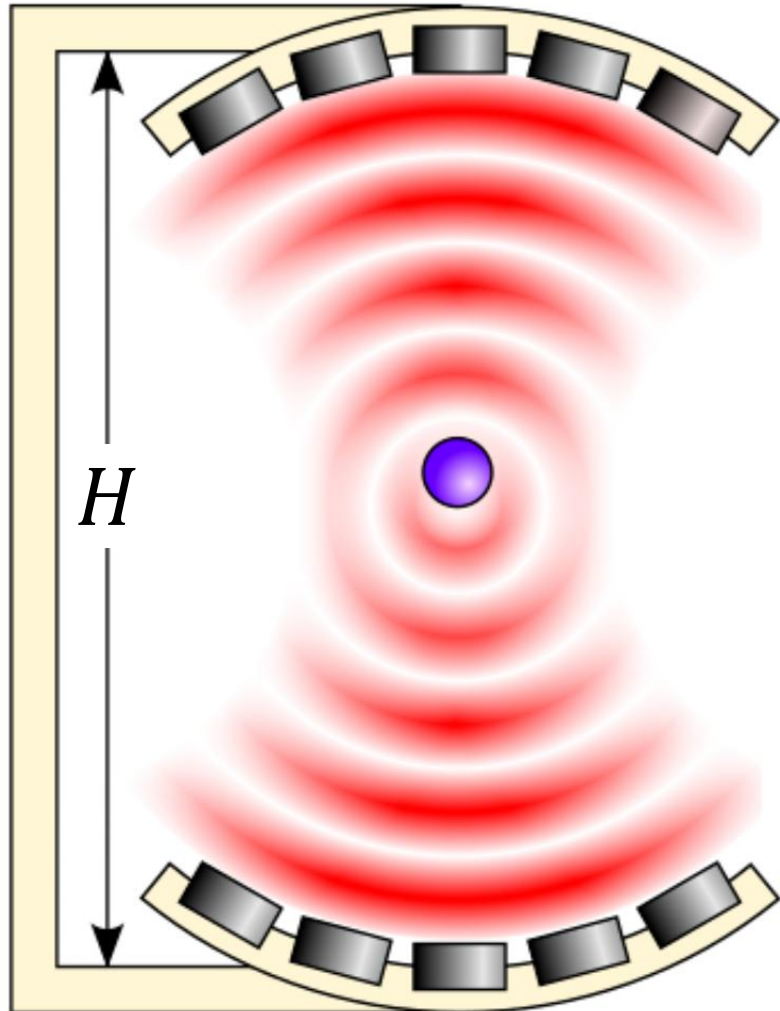


(2) Curved spaces





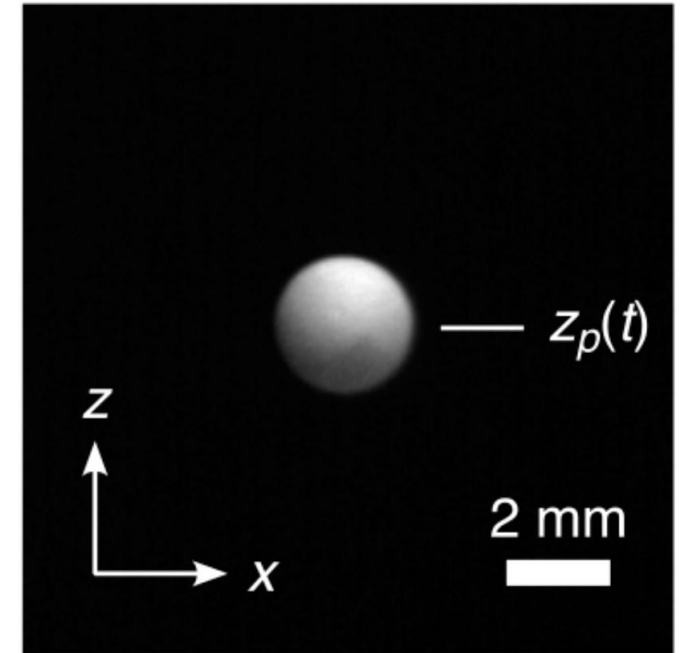
Project 28: Measuring using Sound



Acoustic Trap

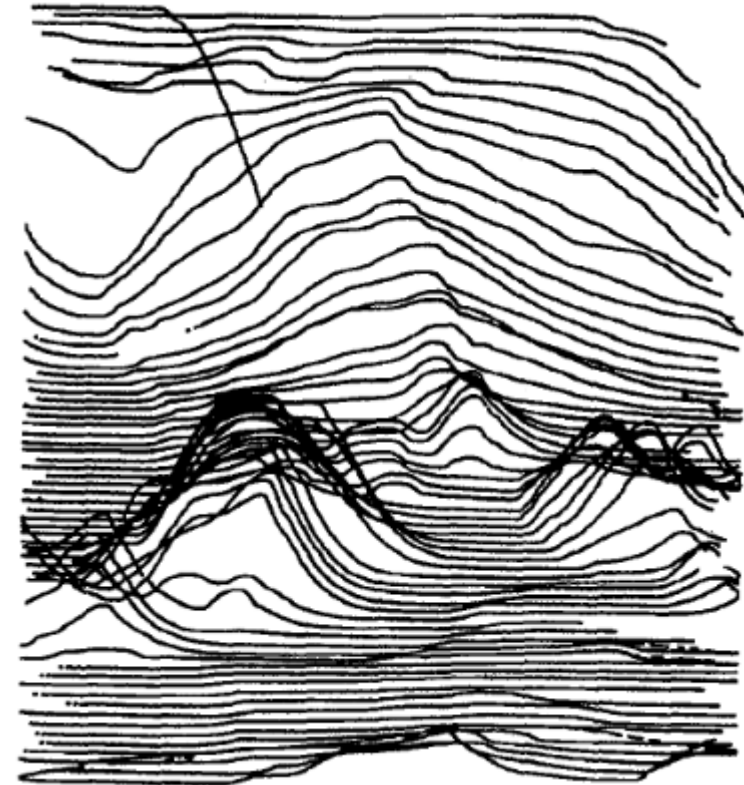
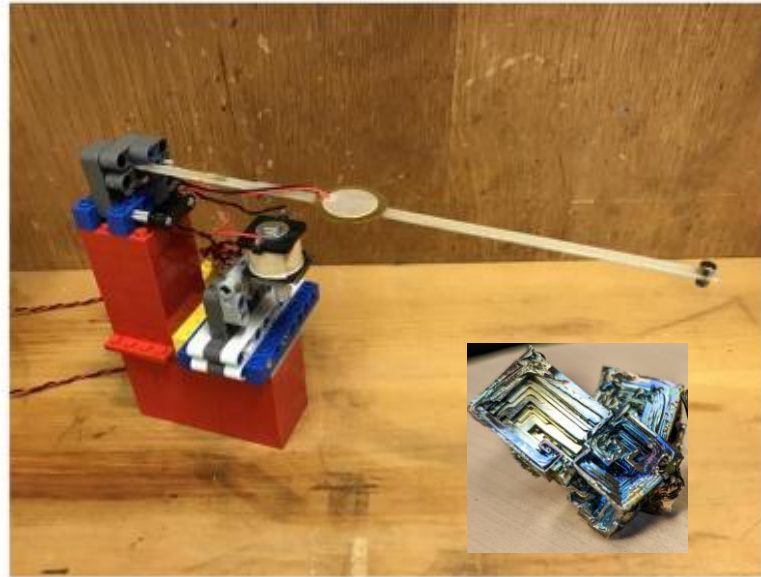
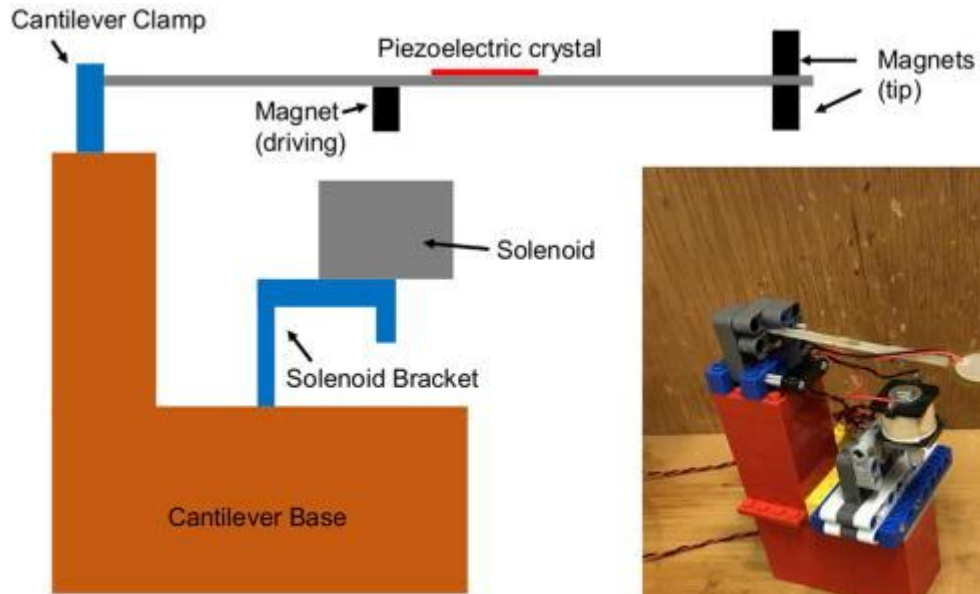
(1) Viscosity

(2) Mass Balance





Project 29: Home built Scanning Probe Microscope



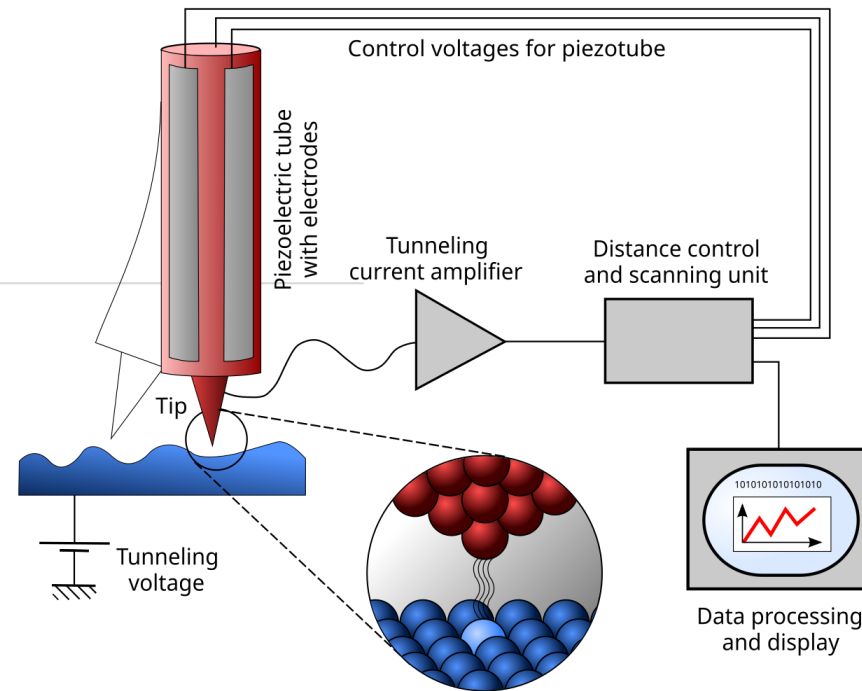
Nadine van der Heijden
n.j.vanderheijden@uu.nl

Project option: one of my projects can be replaced by an open project (please discuss this today!)





Project 30: Looking at atoms with the Nanosurf NaioSTM



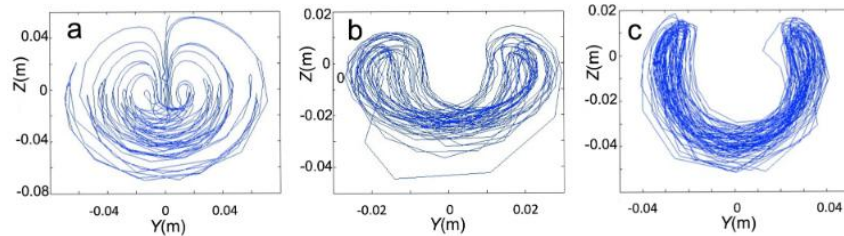
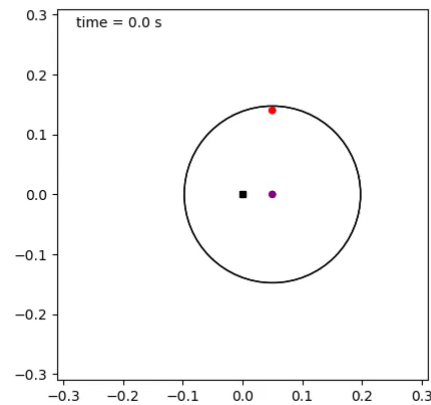
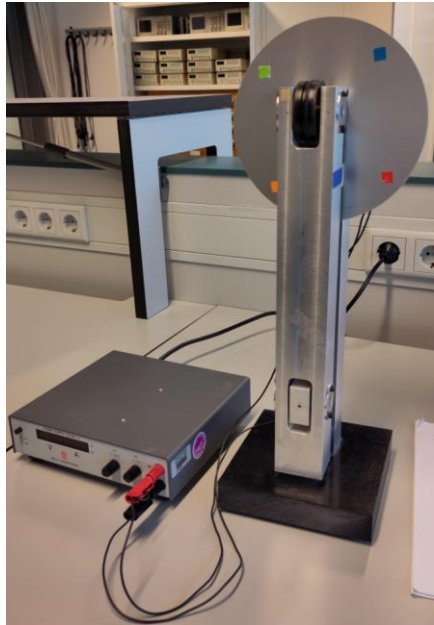
Nadine van der Heijden
n.j.vanderheijden@uu.nl

Project option: one of my projects can be replaced by an open project (please discuss this today!)





Project 31 The upward driven disk



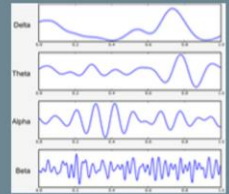
Peter van Capel

p.j.s.vancapel@uu.nl

Project 32 Build your own EEG

1 Introduction

- Building an EEG → device to monitor brain activity
- How to measure significantly small electrical waves from the brain?
- The importance of band-pass filters in neuroresearch



Project option: one of my projects can be replaced by an open project (please discuss this today!)